

Functions and Function Notation (Part II) Homework

The four functions below are from the library. Use the graphs in the library to write the domain and range in words, set notation, and interval notation.

① $f(x) = x^2$ Domain

Words:

Set Notation:

Interval Notation:

Range

Words:

Set Notation:

Interval Notation:

② $f(x) = \frac{1}{x}$ Domain

Words:

Set Notation:

Interval Notation:

Range

Words:

Set Notation:

Interval Notation:

③ $f(x) = \sqrt{x}$

Domain

Words:

Set Notation:

Interval Notation:

Range

Words:

Set Notation:

Interval Notation:

④ $f(x) = x$

Domain

Words:

Set Notation:

Interval Notation:

Range

Words:

Set Notation:

Interval Notation:

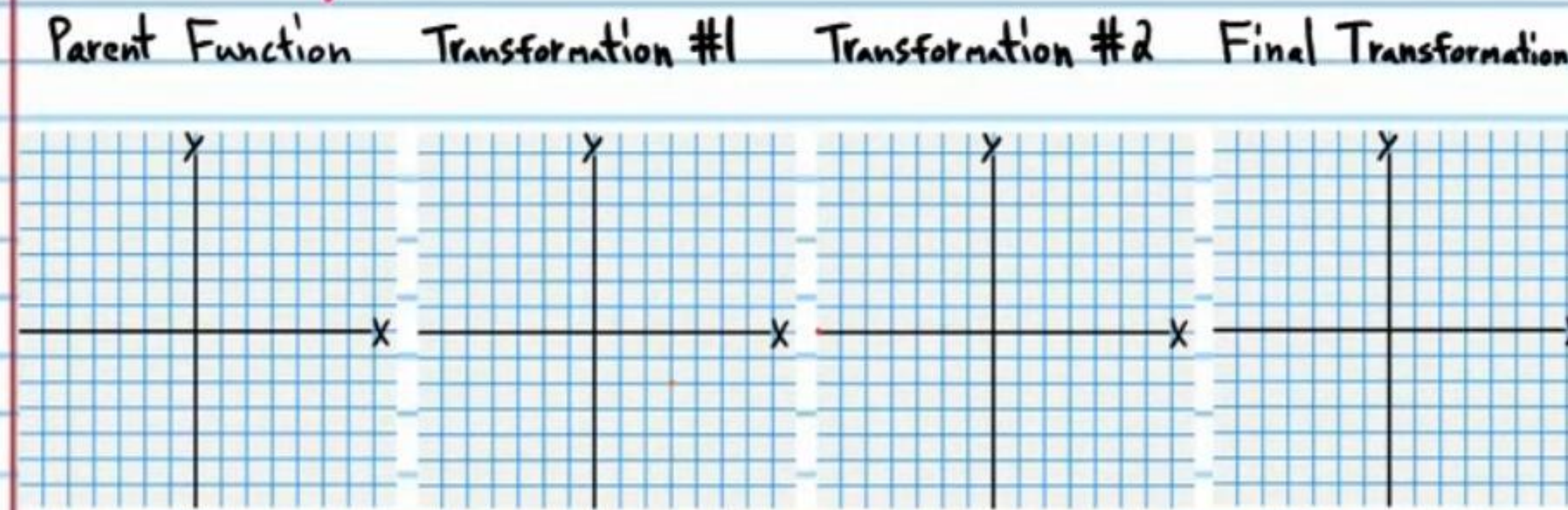
Graph the following functions and use the graphs to determine the domain and range. Write in word form, set notation, and interval notation.

⑤ $f(x) = 3|-x+2|$

↓ Convert $f(x)$ to y .

↓ Change the form of the equation.

↓ Graph the equation.



Domain

Range

Words:

Words:

Set Notation:

Set Notation:

Interval Notation:

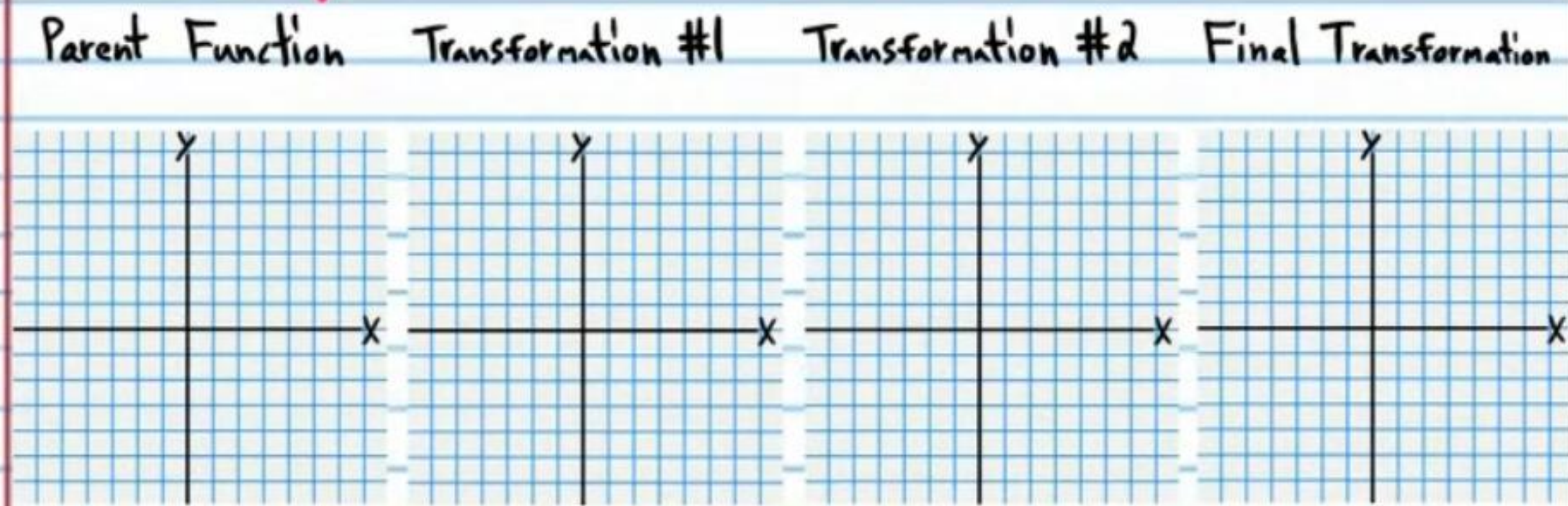
Interval Notation:

⑥ $F(x) = 2^{-3x+1} + 4$

↓ Convert $F(x)$ to y .

↓ Change the form of the equation.

↓ Graph the equation.



Domain

Words:

Set Notation:

Interval Notation:

Range

Words:

Set Notation:

Interval Notation:

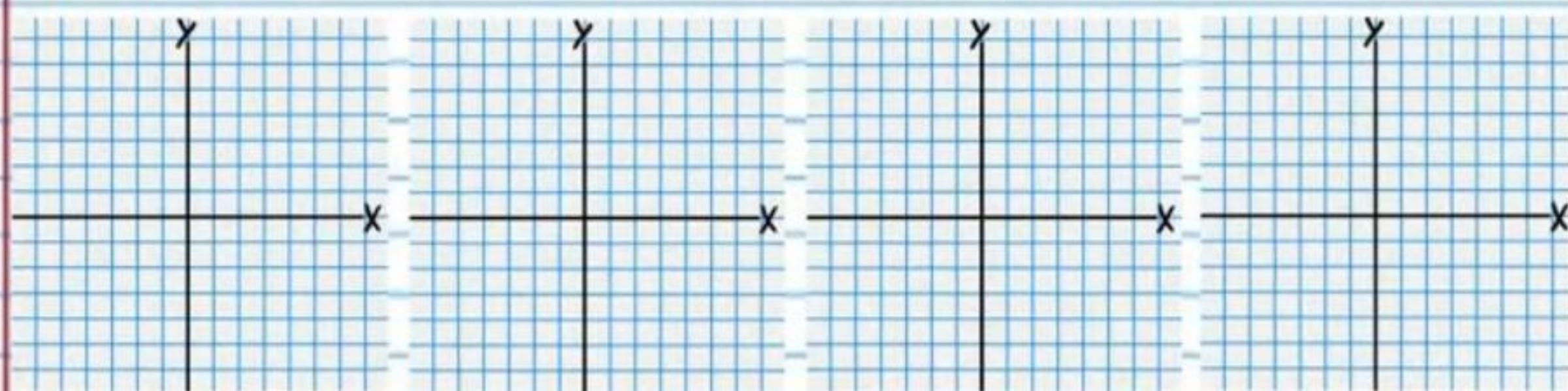
⑦ $f(x) = -2\sqrt{x-3}$

↓ Convert $f(x)$ to y .

↓ Change the form of the equation.

↓ Graph the equation.

Parent Function Transformation #1 Transformation #2 Final Transformation



Domain

Words:

Set Notation:

Interval Notation:

Range

Words:

Set Notation:

Interval Notation:

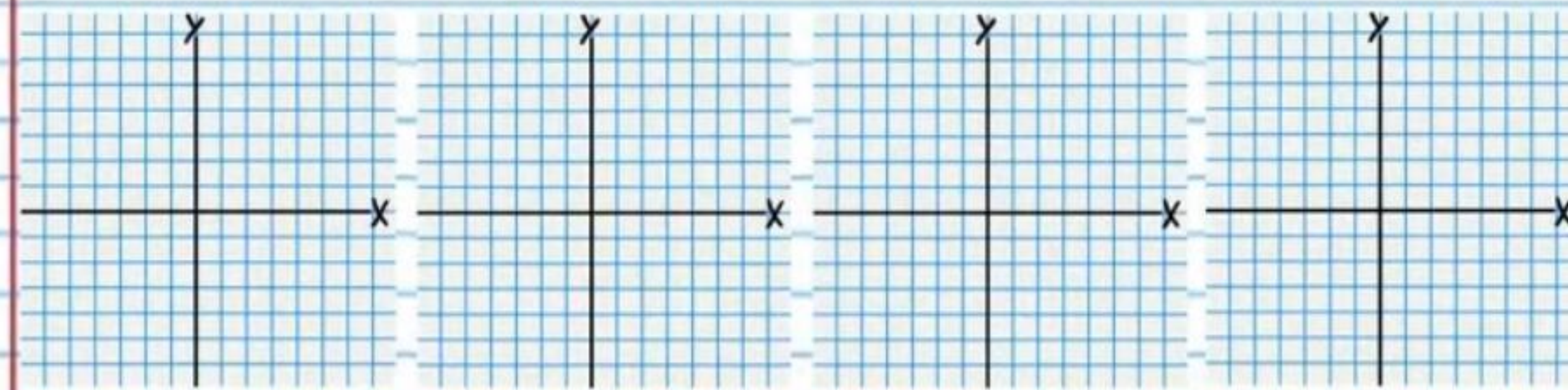
⑧ $f(x) = \frac{(-x-1)^2}{3} + 2$

↓ Convert $f(x)$ to y .

↓ Change the form of the equation.

↓ Graph the equation.

Parent Function Transformation #1 Transformation #2 Final Transformation



Domain

Words:

Set Notation:

Interval Notation:

Range

Words:

Set Notation:

Interval Notation:

* You might have to move the asymptote when transforming.

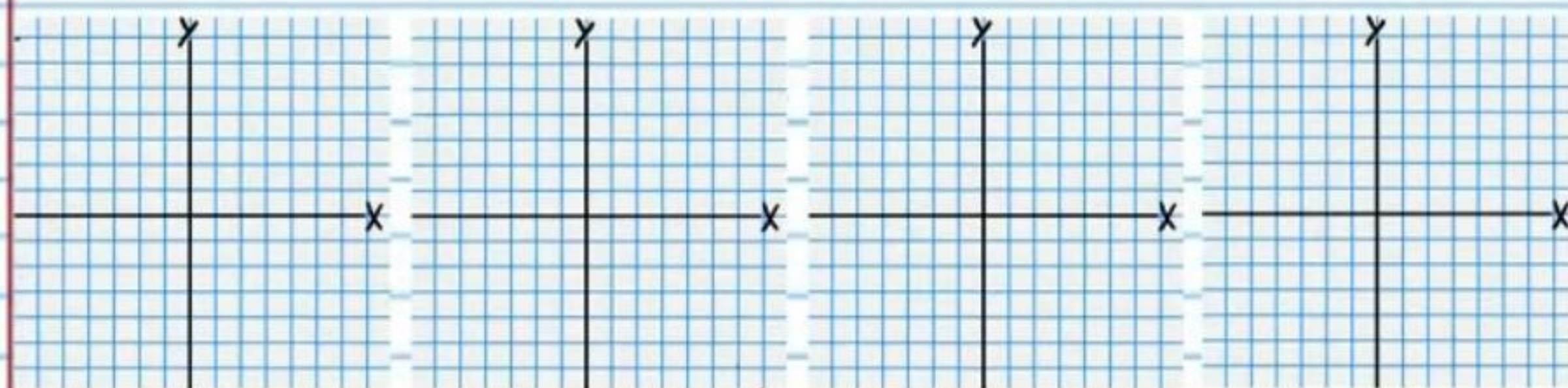
⑨ $f(x) = -\frac{1}{2x-1} + 4$

↓ Convert $f(x)$ to y .

↓ Change the form of the equation.

↓ Graph the equation.

Parent Function Transformation #1 Transformation #2 Final Transformation



Domain

Words:

Set Notation:

Interval Notation:

Range

Words:

Set Notation:

Interval Notation:

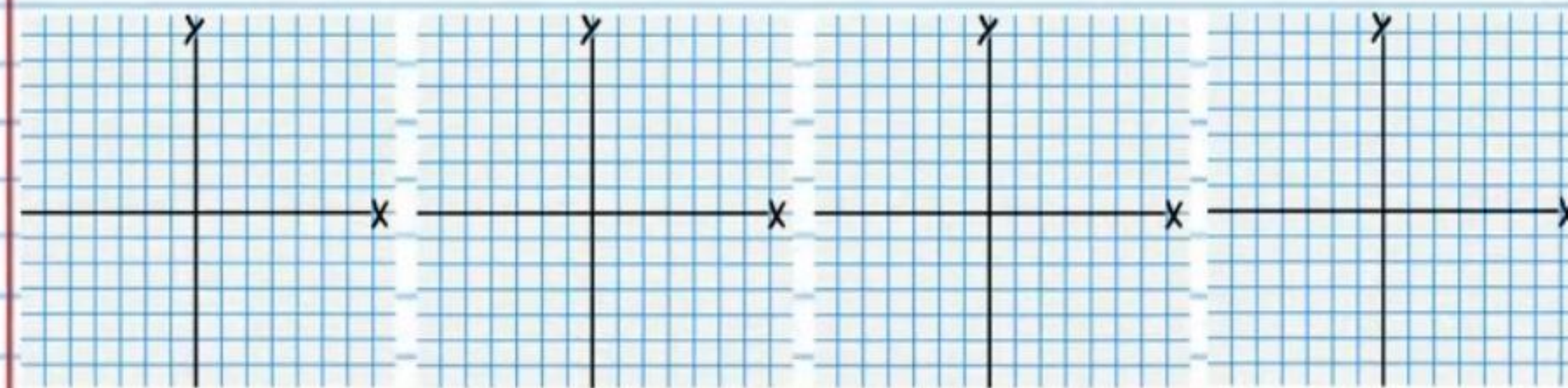
⑩ $f(x) = -2\log_3(x-2) + 1$

↓ Convert $f(x)$ to y .

↓ Change the form of the equation.

↓ Graph the equation.

Parent Function Transformation #1 Transformation #2 Final Transformation



Domain

Words:

Set Notation:

Interval Notation:

Range

Words:

Set Notation:

Interval Notation:

Answers

① $f(x) = x^2$

Domain

Words: All real numbers.

Set Notation: All real numbers.

Interval Notation: $(-\infty, \infty)$

Range

Words: All real numbers greater than or equal to zero.

Set Notation: $\{y | y \geq 0\}$

Interval Notation: $[0, \infty)$

② $f(x) = \frac{1}{x}$

Domain

Words: All real numbers except zero.

Set Notation: $\{x | x \neq 0\}$

Interval Notation: $(-\infty, 0) \cup (0, \infty)$

Range

Words: All real numbers except zero.

Set Notation: $\{y | y \neq 0\}$

Interval Notation: $(-\infty, 0) \cup (0, \infty)$

③ $f(x) = \sqrt{x}$

Domain

Words: All real numbers greater than or equal to zero.

Set Notation: $\{x | x \geq 0\}$

Interval Notation: $[0, \infty)$

Range

Words: All real numbers greater than or equal to zero.

Set Notation: $\{y | y \geq 0\}$

Interval Notation: $[0, \infty)$

④ $f(x) = x$

Domain

Words: All real numbers.

Set Notation: All real numbers.

Interval Notation: $(-\infty, \infty)$

Range

Words: All real numbers.

Set Notation: All real numbers.

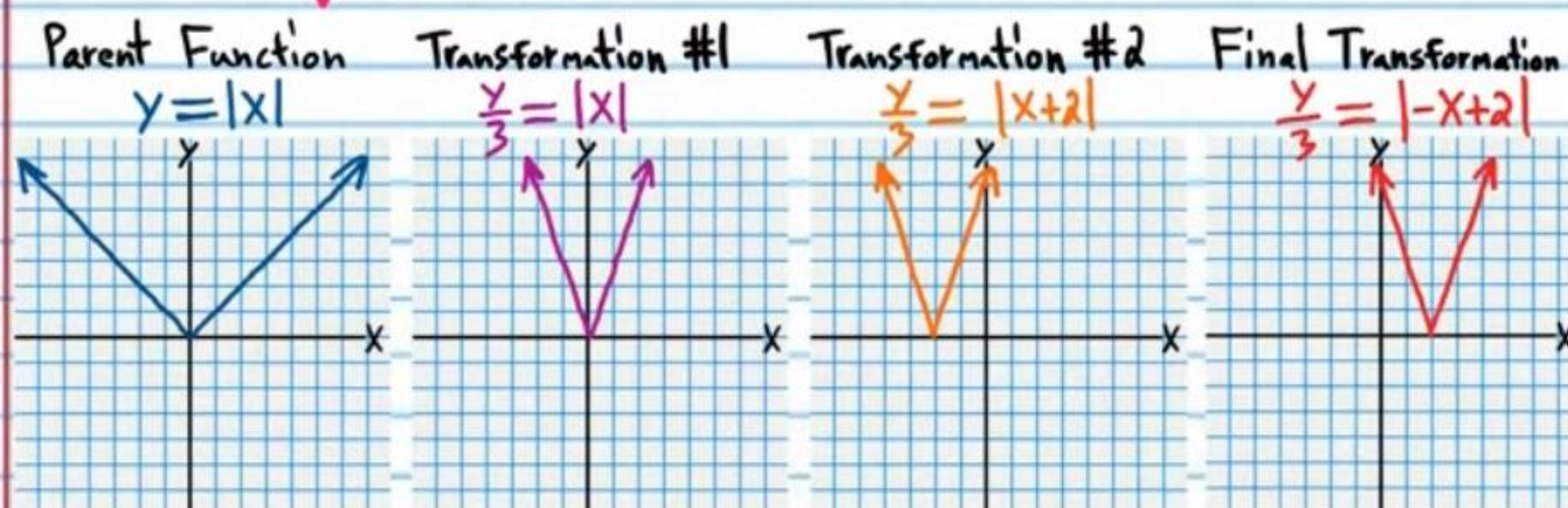
Interval Notation: $(-\infty, \infty)$

⑤ $f(x) = 3|-x+2|$

↓ Convert $f(x)$ to y .
 $y = 3|-x+2|$

↓ Change the form of the equation.
 $\frac{y}{3} = |-x+2|$

↓ Graph the equation.



Domain

Words: All real numbers.

Set Notation: All real numbers.

Interval Notation: $(-\infty, \infty)$

Range

Words: All real numbers greater than or equal to zero.

Set Notation: $\{y | y \geq 0\}$

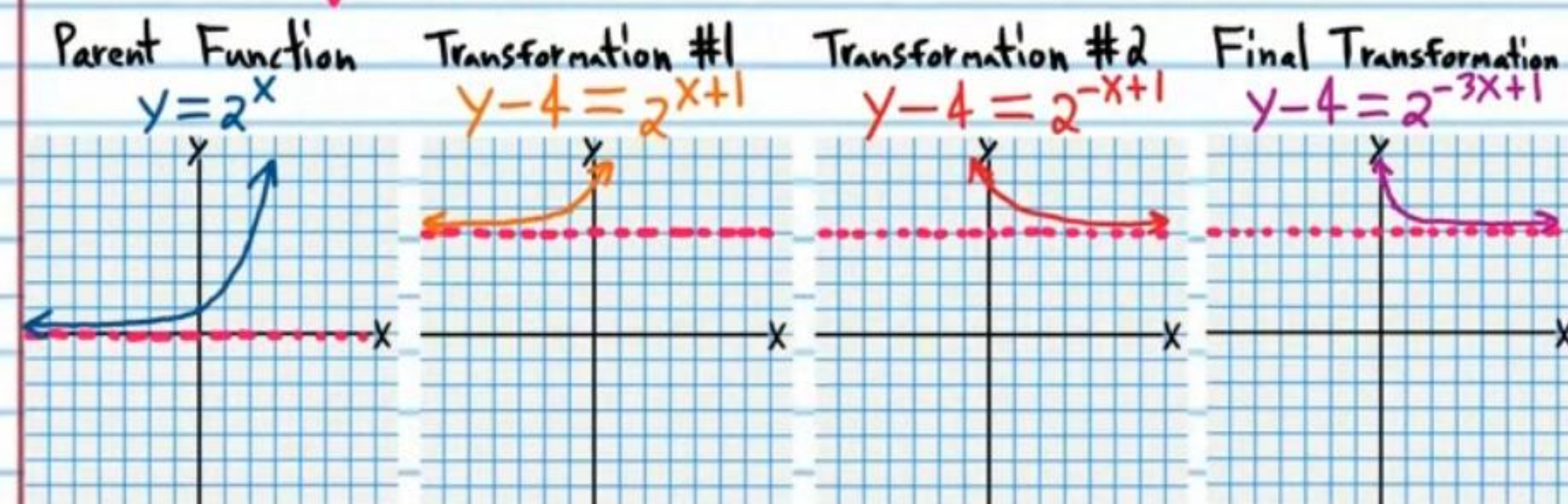
Interval Notation: $[0, \infty)$

⑥ $f(x) = 2^{-3x+1} + 4$

↓ Convert $f(x)$ to y .
 $y = 2^{-3x+1} + 4$

↓ Change the form of the equation.
 $y-4 = 2^{-3x+1}$

↓ Graph the equation.



Domain

Words: All real numbers.

Set Notation: All real numbers.

Interval Notation: $(-\infty, \infty)$

Range

Words: All real numbers greater than 4.

Set Notation: $\{y | y > 4\}$

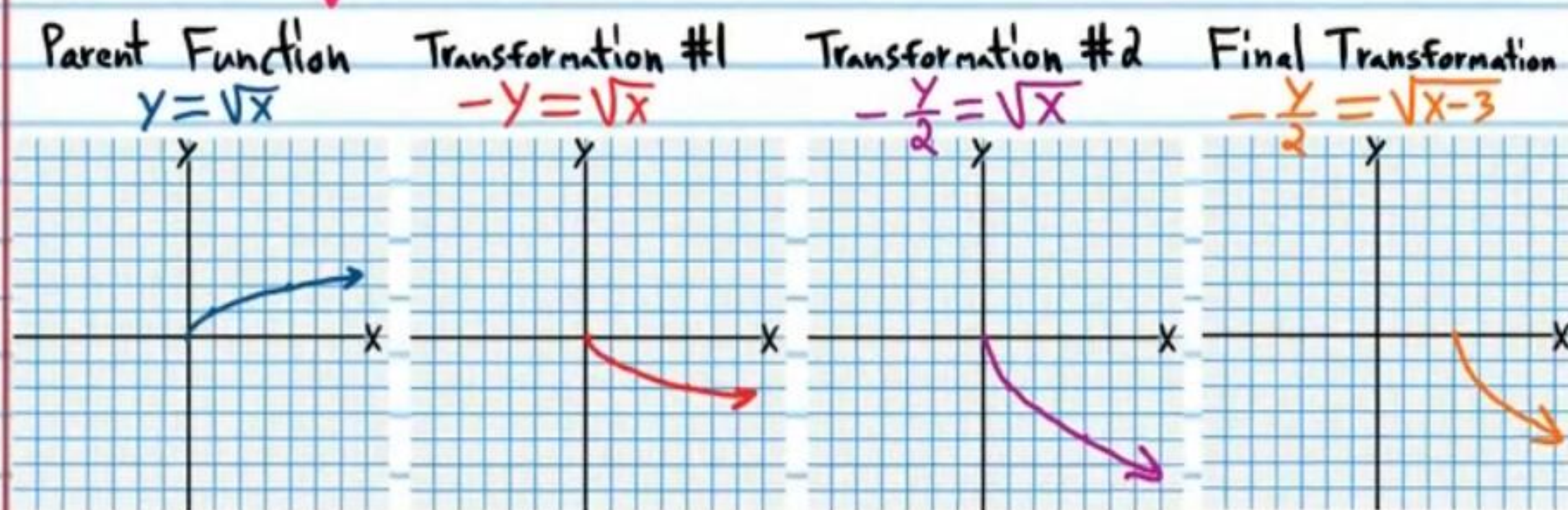
Interval Notation: $(4, \infty)$

⑦ $f(x) = -2\sqrt{x-3}$

↓ Convert $f(x)$ to y .
 $y = -2\sqrt{x-3}$

↓ Change the form of the equation.
 $-\frac{y}{2} = \sqrt{x-3}$

↓ Graph the equation.



Domain: Words: All real numbers greater than or equal to 3.

Set Notation: $\{x \mid x \geq 3\}$

Interval Notation: $[3, \infty)$
 Range

Words: All real numbers less than or equal to zero.

Set Notation: $\{y \mid y \leq 0\}$

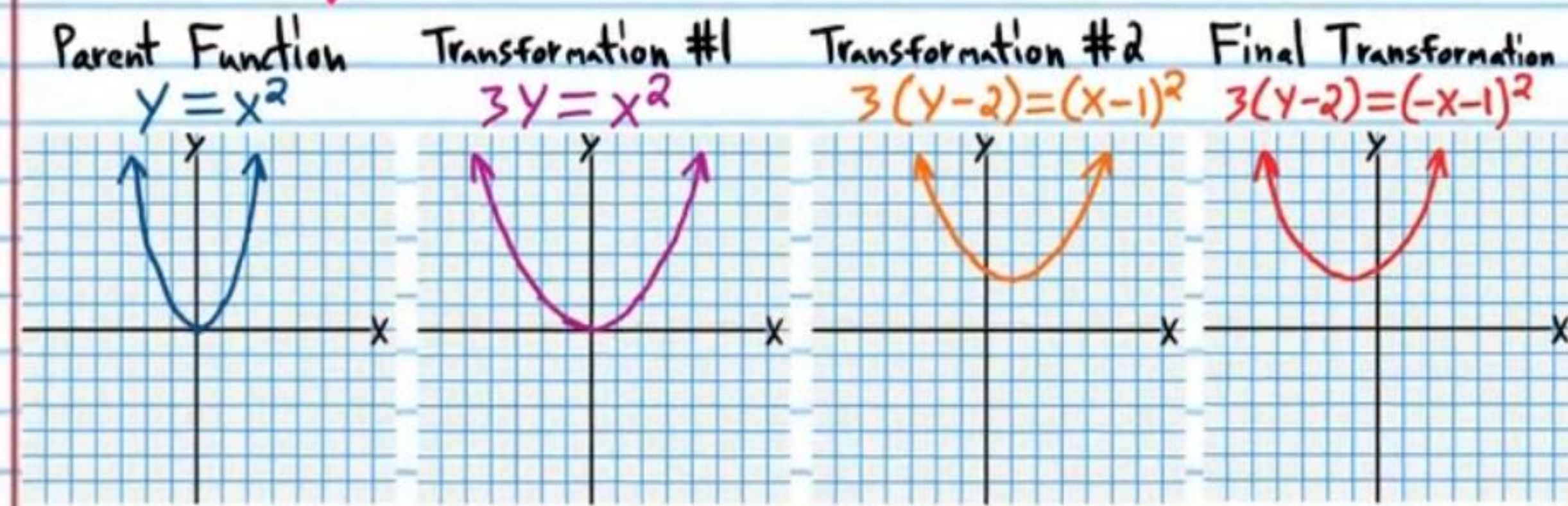
Interval Notation: $(-\infty, 0]$

⑧ $f(x) = \frac{(-x-1)^2}{3} + 2$

↓ Convert $f(x)$ to y .
 $y = \frac{(-x-1)^2}{3} + 2$

↓ Change the form of the equation.
 $3(y-2) = (-x-1)^2$

↓ Graph the equation.



Domain: Words: All real numbers.

Set Notation: All real numbers.

Interval Notation: $(-\infty, \infty)$
 Range

Words: All real numbers greater than or equal to 2.

Set Notation: $\{y \mid y \geq 2\}$

Interval Notation: $[2, \infty)$

$$⑨ \quad f(x) = -\frac{1}{2x-1} + 4$$

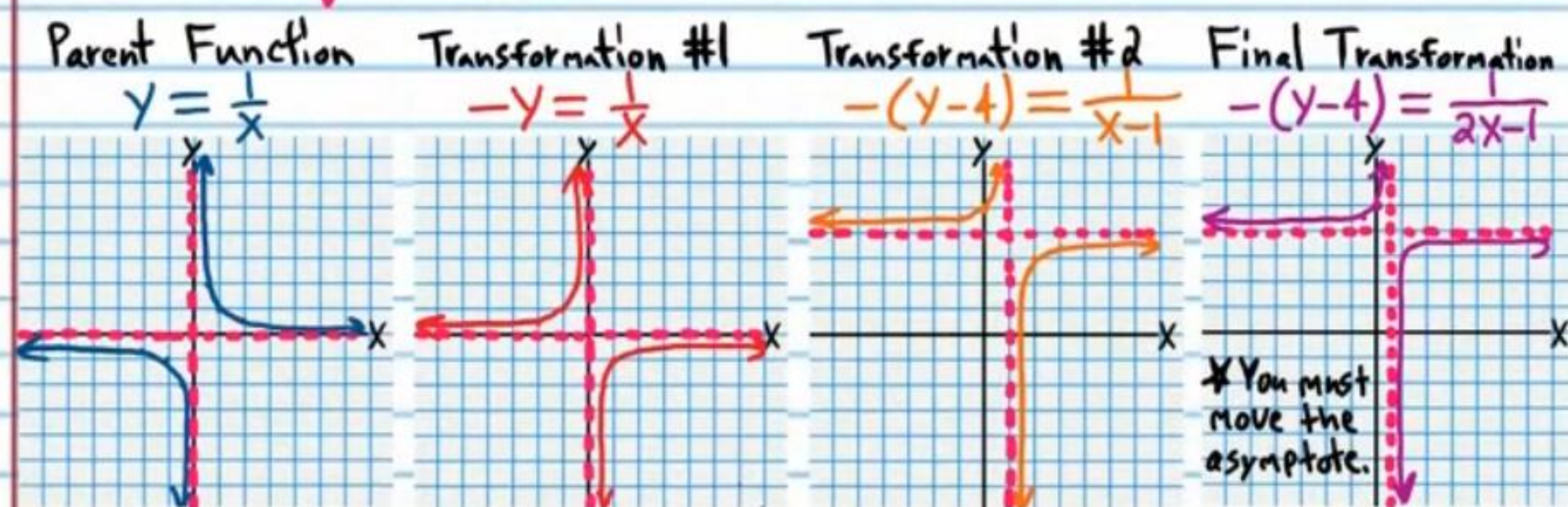
Convert $f(x)$ to y .

$$y = -\frac{1}{2x-1} + 4$$

Change the form of the equation.

$$-(y-4) = \frac{1}{2x-1}$$

Graph the equation.



Domain

Words: All real numbers except $\frac{1}{2}$.

Set Notation: $\{x \mid x \neq \frac{1}{2}\}$

Interval Notation: $(-\infty, \frac{1}{2}) \cup (\frac{1}{2}, \infty)$

Range

Words: All real numbers except 4.

Set Notation: $\{y \mid y \neq 4\}$

Interval Notation: $(-\infty, 4) \cup (4, \infty)$

$$⑩ \quad f(x) = -2\log_3(x-2) + 1$$

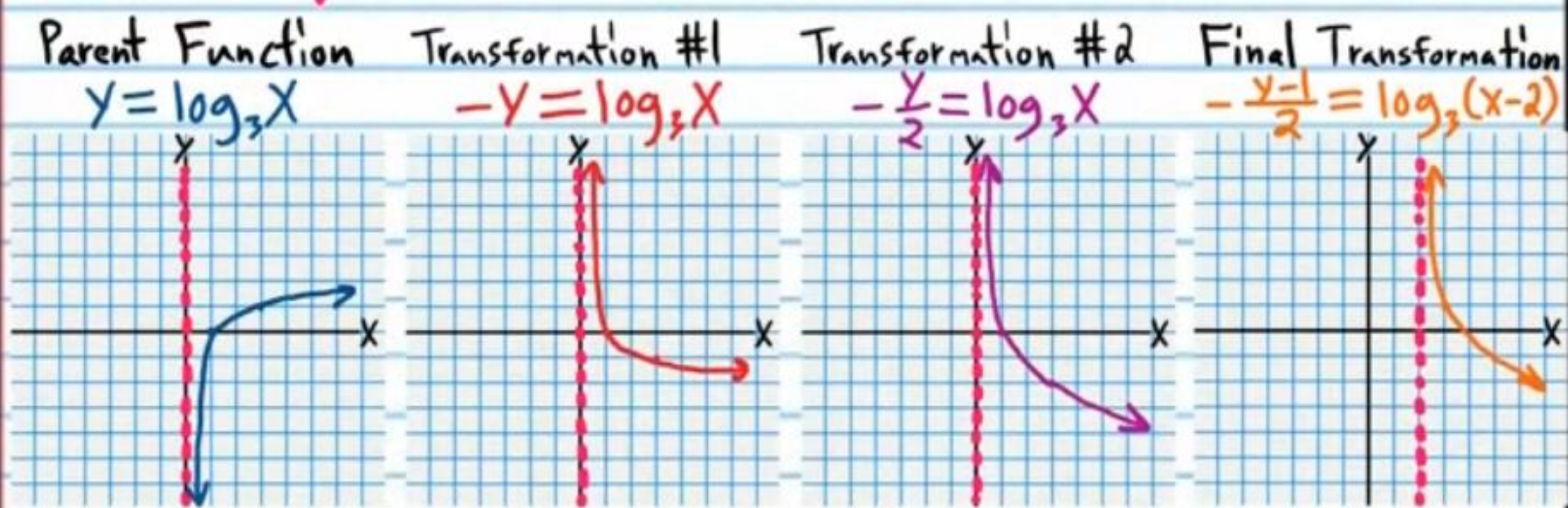
Convert $f(x)$ to y .

$$y = -2\log_3(x-2) + 1$$

Change the form of the equation.

$$-\frac{y-1}{2} = \log_3(x-2)$$

Graph the equation.



Domain

Words: All real numbers greater than 2.

Set Notation: $\{x \mid x > 2\}$

Interval Notation: $(2, \infty)$

Range

Words: All real numbers.

Set Notation: All real numbers.

Interval Notation: $(-\infty, \infty)$